

Case Study No: 4

Experiment Title: To study the functions, scopes and closures in JavaScript.

Case Study Title: To build a Palindrome Checker.

Software/ Tools Required: VS Code (text editor), HTML, CSS, JavaScript.

Case Study Program Code

Index.html –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&family=Playfair+Display:wght@700&display=swap"
rel="stylesheet">
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Palindrome Checker</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
  <h1>Palindrome Checker</h1>
  <p>Discover whether your text is a palindrome.</p>
  <input type="text" id="textInput" placeholder="Enter text here">
  <button onclick="checkPalindrome()">Check</button>
  <div id="result"></div>
</div>
<div>
  <script src="script.js"></script>
</div>
</body>
</html>
```

style.css –

```
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
}

body{
  display:flex;
  justify-content:center;
  align-items:center;
  min-height:100vh;
```

```
font-family:'Poppins',sans-serif;
background:linear-gradient(135deg,#FFF8F2,#F4F0FF,#EDF8FF);
padding:20px;
}
```

```
.container{
width:420px;
background:rgba(255,255,255,0.75);
backdrop-filter:blur(18px);
-webkit-backdrop-filter:blur(18px);
border:1px solid rgba(255,255,255,.6);
border-radius:25px;
padding:45px 35px;
text-align:center;
box-shadow:0 20px 45px rgba(60,60,60,.12);
transition:.3s;
}
```

```
.container:hover{
transform:translateY(-4px);
box-shadow:0 25px 55px rgba(60,60,60,.15);
}
```

```
h1{
font-family:'Playfair Display',serif;
font-size:42px;
color:#5C4B99;
margin-bottom:12px;
letter-spacing:1px;
}
```

```
p{
color:#777;
font-size:15px;
margin-bottom:28px;
line-height:1.6;
}
```

```
input{
width:100%;
padding:15px;
font-size:16px;
border:2px solid #E7E2FA;
border-radius:12px;
background:#FCFCFF;
outline:none;
transition:.3s;
}
```

```
input:focus{
border-color:#8B7CF7;
box-shadow:0 0 12px rgba(139,124,247,.2);
}
```

```
button{
width:100%;
margin-top:18px;
```

```

padding:15px;
border:none;
border-radius:12px;
background:#8B7CF7;
color:white;
font-size:17px;
font-weight:600;
cursor:pointer;
transition:.3s;
}

button:hover{
background:#6F61E8;
transform:translateY(-2px);
}

button:active{
transform:scale(.98);
}

#result{
margin-top:25px;
font-size:22px;
font-weight:600;
min-height:30px;
}

@media(max-width:480px){
.container{
width:100%;
padding:35px 25px;
}
h1{
font-size:34px;
}
}

```

script.js –

```

// Function Declaration
function reverseString(text) {
return text.split("").reverse().join("");
}

// Closure
function palindromeChecker() {
let count = 0;
return function(text){
count++;
const cleaned = text
.toLowerCase()
.replace(/[^a-z0-9]/g, "");
return{
isPalindrome: cleaned === reverseString(cleaned),
checks: count
}
}
}

```

```

    };
  }
}

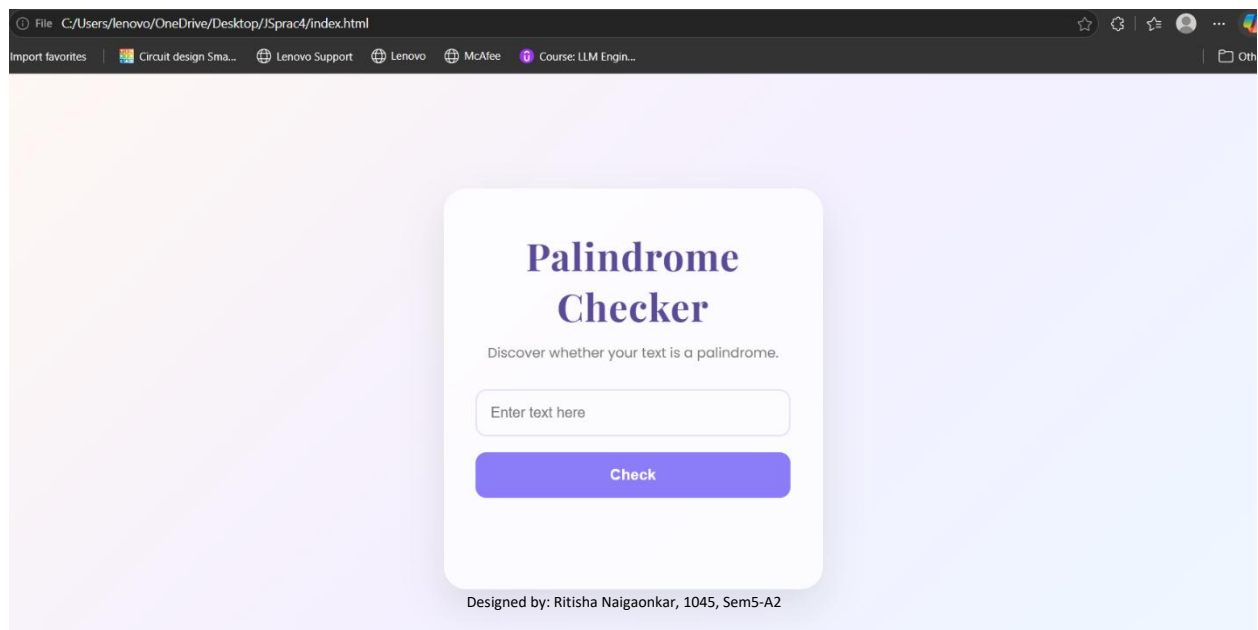
const check = palindromeChecker();

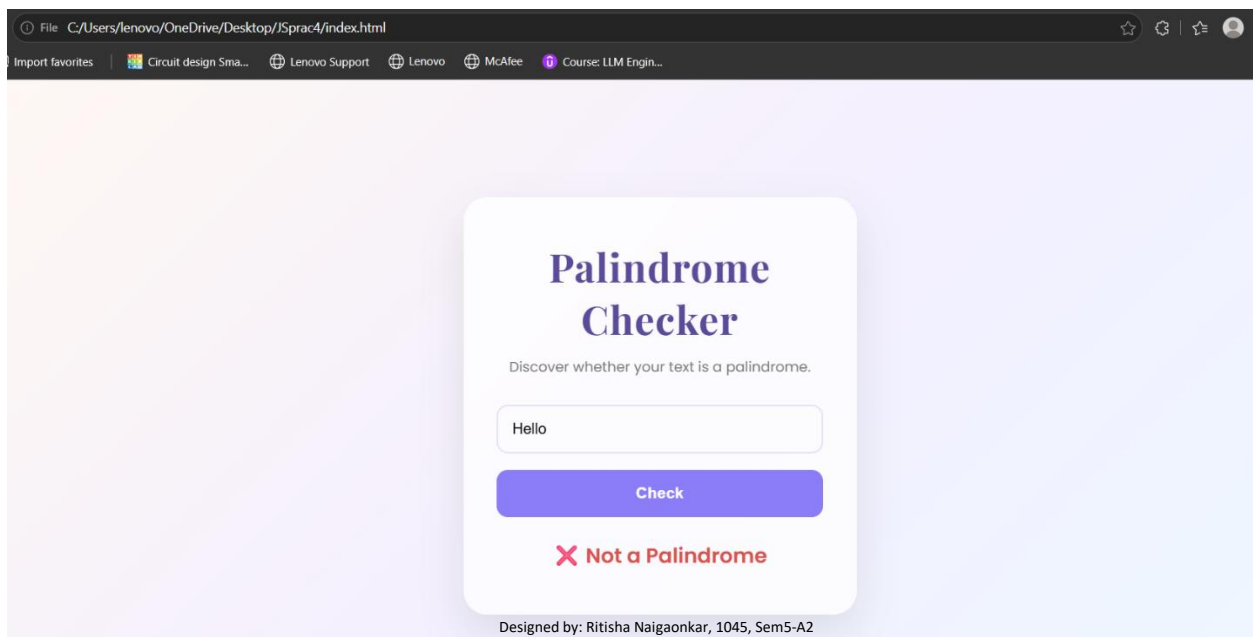
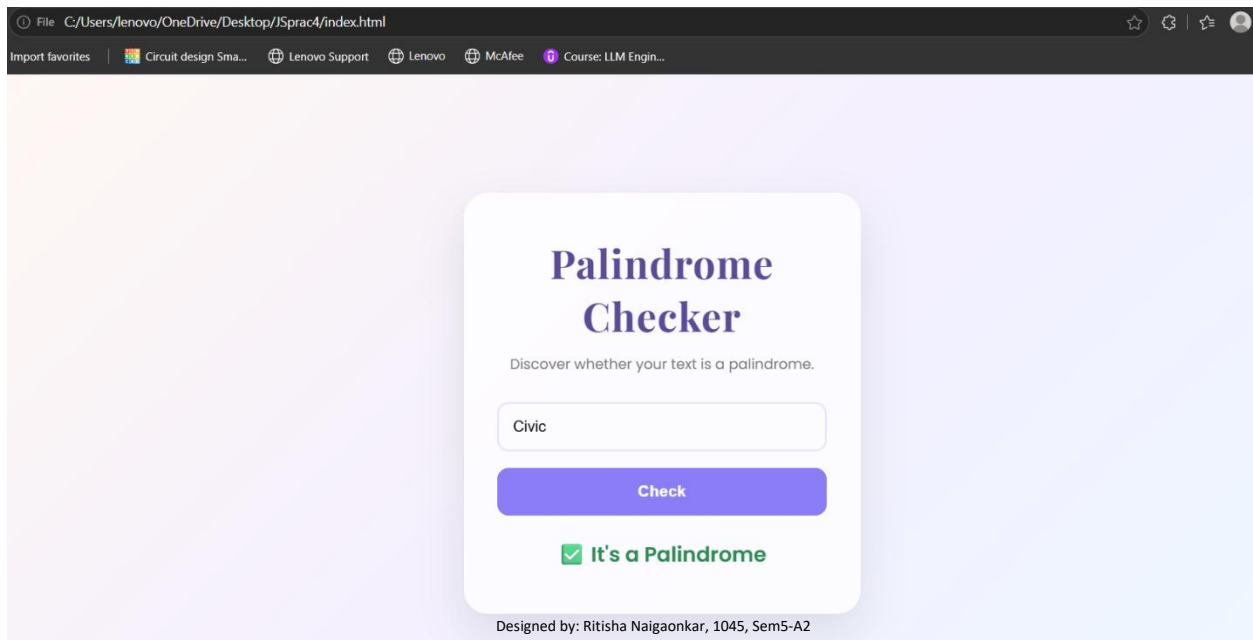
// Arrow Function
const showResult = (msg,color)=>{
  const result=document.getElementById("result");
  result.innerHTML=msg;
  result.style.color=color;
}

function checkPalindrome(){
  try{
    const input=document.getElementById("textInput").value;
    if(input.trim()=="")
      throw "Please enter some text.";
    const output=check(input);
    if(output.isPalindrome){
      showResult("\u2705 It's a Palindrome", "#2e8b57");
    }
    else{
      showResult("\u274c Not a Palindrome", "#d9534f");
    }
  }
  catch(error){
    showResult(error, "#ff9800");
  }
}

```

Output:





Result/ Conclusion: Thus a JavaScript program to check whether a given string is a Palindrome was successfully implemented, using function declarations, closures, and arrow functions, and the desired output was verified.

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